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GCHN-DTI: Predicting drug-target interactions by graph convolution on heterogeneous networks

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ABSTRACT

Determining the interaction of drug and target plays a key role in the process of drug development and discovery. The calculation methods can predict new interactions and speed up the process of drug development. In recent studies, the network-based approaches have been proposed to predict drug-target interactions. However, these methods cannot fully utilize the node information from heterogeneous networks. Therefore, we propose a method based on heterogeneous graph convolutional neural network for drug-target interaction prediction, GCHN-DTI (Predicting drug-target interactions by graph convolution on heterogeneous net-works), to predict potential DTIs. GCHN-DTI integrates network information from drug-target interactions, drug-drug interactions, drug-similarities, target-target interactions, and target-similarities. Then, the graph convolution operation is used in the heterogeneous network to obtain the node embedding of the drugs and the targets. Furthermore, we incorporate an attention mechanism between graph convolutional layers to combine node embedding from each layer. Finally, the drug-target interaction score is predicted based on the node embedding of the drugs and the targets. Our model uses fewer network types and achieves higher prediction performance. In addition, the prediction performance of the model will be significantly improved on the dataset with a higher proportion of positive samples. The experimental evaluations show that GCHN-DTI outperforms several state-of-the-art prediction methods.

1. Introduction

The prediction of drug-target interactions is of great significance for drug repositioning [1–3], drug discovery [4,5], side-effect prediction [6] and drug resistance [7]. The wet experiment method predicts the interaction between drug and target, which is expensive and cumbersome [8,9]. Identifying drug-target interactions (DTIs) based on computational approaches can greatly narrow down the large search space of drug candidates for downstream experimental validation, and thus significantly reduce the high cost and the long period of developing a new drug [10].

The traditional DTIs prediction approaches are mainly categorized into docking-based approaches [11,12] and ligand-based approaches [13]. However, these structure-based methods are usually limited by the target proteins of three-dimensional structural information. In addition, some ligand-based method usually requires a large amount of drug-target interactions data. Due to insufficient data on the interactions between proteins and ligands, the performance of many methods has been restricted.

In recent years, the network-based approaches have demonstrated great advantages compared to those docking-based and ligand-based methods [14–17]. A key idea of the network-based approach is the assumption that similar drugs may share similar targets [18]. The network-based method formulates the prediction of DTIs as link prediction problem in a graph. The network-based method mainly consists of two steps: network construction and DTI identification. In order to identify the new DTIs, the network-based method calculates the similarity between drugs and targets according to the network topology, and

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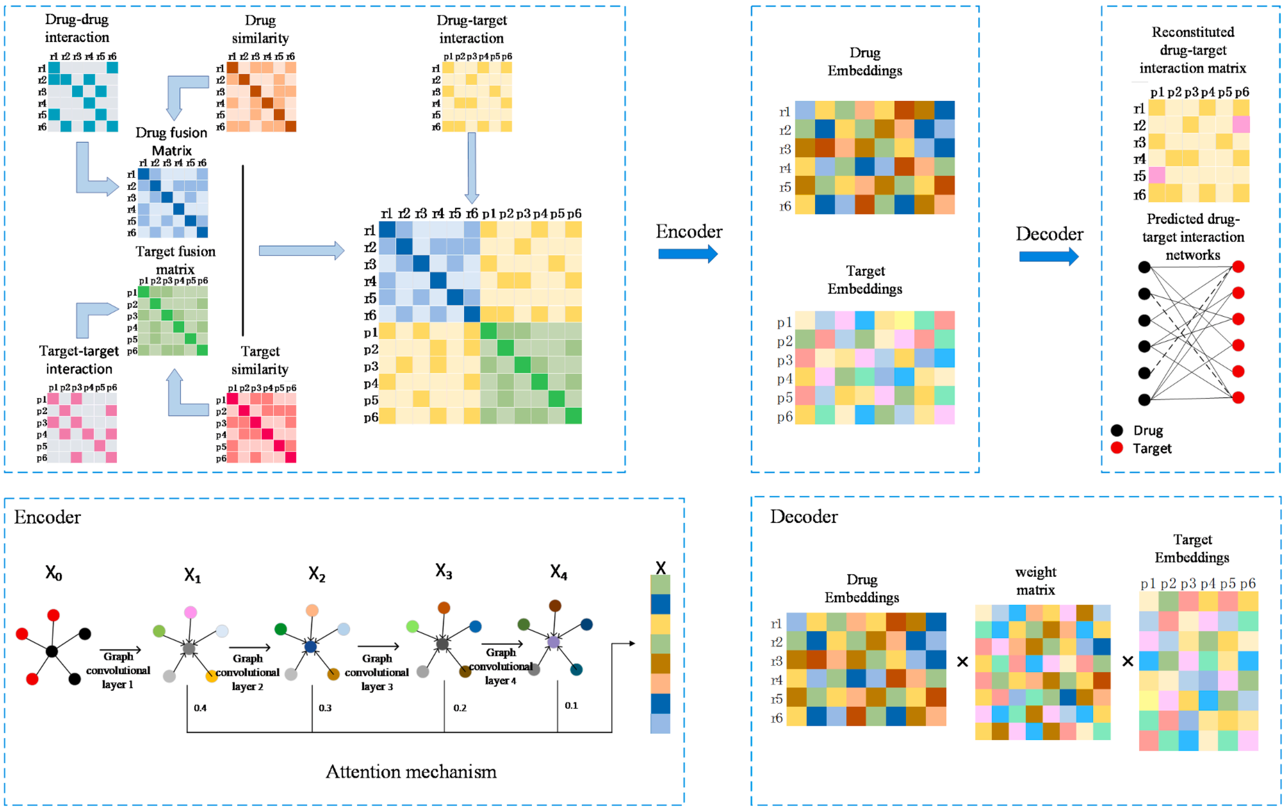

Fig. 1. GCHN-DTI pipeline flowchart. First, the drug network and the target network are fused by linear combination to obtain the drug fusion network and the target fusion network, and then the drug fusion network, the target fusion network and the drug-target interaction network are constructed as a heterogeneous network. Then, the heterogeneous network containing drug information and protein information is fed into the encoder of the graph convolutional network to obtain low-dimensional representations of drugs and proteins. Finally, the trainable weight matrix, drug low-dimensional matrix, and protein low-dimensional matrix are input into the decoder to obtain the reconstructed interaction matrix. Each position of the reconstructed interaction matrix represents a new drug-target interaction score.

builds a network that includes drugs, proteins, or more types of nodes. Bipartite graph is the most common network structure in the network-based methods, drugs and proteins are two different types of nodes in the network, the edges between drugs and proteins represent known interactions. The purpose of network-based method is to predict unknown interactions based on known interactions.

Yildirim [19] constructed a network structure of drugs and targets, used computer algorithms to process drug-target network data, and calculated the interaction between drug com-pounds and protein sequences. The experiment found that a drug molecule with drug effect interacts with multiple target proteins, and similarly, a target protein also interacts with multiple drugs. The bipartite local model with neighbor-based interaction-profile inferring (BLMNII) [20] is a combination of the bipartite local model (BLM) and the neighbor-based interaction-profile inferring (NII). The BLM framework models the drug-target interactions prediction task as a binary classification problem in a bipartite graph. BLM labels each known drug by whether it interacts with target or not, and then trains an SVM based on the drug similarity matrix. Similarly, BLM also labels each known target by whether it interacts with drug or not, and then trains an SVM based on the target similarity matrix. The final prediction of whether drug interacts with target is then derived based on the average prediction score from both directions. The NII procedure incorporates the neighbors' interaction profiles into the BLM method to train the model and enable the prediction for new drugs or targets.

Luo [1] provided a prediction method based on heterogeneous networks, which extracted the low-dimensional characteristic information from heterogeneous data sources and used the inductive matrix completion (IMC) approach to predict the drug-target interaction fraction. Wan [21] provided a prediction method called NeoDTI, that integrated neighborhood information of the heterogeneous networks from diverse data sources via a number of information passing and aggregation operations, and NeoDTI applied a network topology preserving learning procedure to enforce the extracted feature representations of drugs and targets to match the observed net-works. DDR [22] is a prediction method that utilized graph mining and machine learning techniques, which is based on heterogeneous graph containing information for the known DTIs, as well as similarities between drugs and similarities between target proteins obtained from different data sources. Zhao [23] proposed the concept of drug–protein pairs (DPPs) in their model, which uses graph convolutional neural networks to extract features from drug-protein pair networks, then, deep neural network (DNN) is used to predict the labels of the drug-protein pairs. The DTI-CDF [24] method extracts features from the heterogeneous DTI weighted graph as an input feature vector of the CDF (cascade deep forest)-based model. The DTI-MLCD [25] method is a multi-label classification framework. It transforms the DTI prediction problem from traditional binary classification into multi-label classification, and introduces the community detection method with the label correlations considered.

In this study, we propose a model that performs graph convolution operations on heterogeneous networks to predict drug-target interactions. Firstly, the heterogeneous network is constructed through the known drug-target interaction matrix, drug similarity matrix, target similarity matrix, drug interaction matrix, and target interaction matrix. Then the graph convolution is performed on the heterogeneous networks to learn the node embedding of drugs and targets. And we introduce an attention mechanism between the convolutional layers to

Table 1
Experimental datasets.

Dataset	Drug	Protein	Associations
Dataset1	708	1512	1923
Dataset2	210	204	1476

Table 2
Number of interacting nodes in Dataset1.

Edge Type	Number
Drug-Protein	1923
Drug-Drug	10,036
Protein-Protein	7363

assign the weights of each layer. Finally, the interaction probability of each drug-target pair is predicted by the score function. The experimentally validated results show that our model achieves AUC of 0.93 on dataset 1 and 0.97 on dataset 2. And we predicted some new drug-target interactions, which is confirmed by other studies. This confirms the validity of the model in identifying drug-target interactions.

2. Methods

2.1. Overview

We propose a method based on graph convolutional neural networks to predict drug-target interactions. First, we prepare five initial networks in the dataset. In order to obtain the sufficient drug nodes and target nodes information, we merged the drug-drug (target-target) interaction network and the drug-drug (target-target) similarity network through the method of network fusion, the fused network is called drug-drug fusion network and target-target fusion network. Then, the drug-drug fusion network, target-target fusion network and drug-target interaction network are constructed into a heterogeneous network, and the graph convolution is performed on the heterogeneous network. The graph convolution operation first needs to obtain the low-dimensional representation of the drug nodes and the target nodes through the encoder. The data processing on low-dimensional vectors is simpler and more effective than high-dimensional vectors [26,27]. In addition, we also introduce an attention mechanism to integrate node information from multiple convolutional layers, and obtain the integrated node low-dimensional representation. Then the low-dimensional representation into the decoding layer is input to reconstruct the drug-target interaction matrix to obtain the specific drug-target interaction probability score, and to predict identify potential drug-target interaction pairs. Fig. 1 shows the overview of our proposed framework.

2.2. Dataset

The dataset in the experiment comes from Luo [1], which imported the known DTIs and drug-drug interactions from DrugBank (Version 3.0) [28]. The protein–protein interactions were downloaded from the HPRD database (Release 9) [29]. The statistical results of the data show that there are 708 drugs and 1,512 targets, and contain 1,923 known drug-target inter-action pairs, 1,068,573 unknown drug-target interaction pairs, 10,036 drug-drug interaction pairs, and 7,363 target-target interaction pairs. In addition, the dataset also includes drug similarity network and target similarity network. In the drug chemical structure similarity network, the similarity score between a pair of two drugs is calculated using the Tanimoto coefficient [11]. In the target protein sequence similarity network, the similarity score between a pair of two proteins is computed using the Smith-Waterman [30] score based on their primary sequences. The ratio of positive and negative samples is 1:556. Table 1 and Table 2 summarize the details of the dataset.

In addition, we had constructed an independent test set. This dataset is called the gold standard data set and is collected by Yamanishi [7]. It has also been used many times in recent studies. which includes four DTIs subsets classified by the types of target proteins (in human): Enzymes (E, including 445 drugs and 664 proteins), Ion Channels (IC, 210 drugs and 204 proteins), G-protein-coupled Receptors (GPCR, 223 drugs and 95 proteins), and Nuclear Receptors (NR, 54 drugs and 26 targets), respectively. We used the IC in it as an independent test set. The ratio of positive and negative samples is 1:29.

2.3. Network fusion

In order to make full utilize of the information of multiple networks, we use a method of linear combination to integrate the drug-drug interaction network with the drug similarity network, and the target-target interaction network with the target similarity network. The core idea of this method is that these known drug-drug pair and target-target pair usually have higher similarity of weight than these unknown interaction drug-drug pairs and target-target pairs. The calculation formula for network fusion is as follows:

$$U^d = W_1 S^d + A^d \tag{1}$$

$$P^p = W_2 S^p + A^p \tag{2}$$

Here W_1 and W_2 are weight parameter, S^d and S^p represent drug similarity matrix and target similarity matrix, A^d and A^p represent drug-drug interaction matrix and target-target interaction matrix, U and P are fusions after the drug matrix and target matrix.

2.4. Heterogeneous network

The heterogeneous network is constructed based on the drug-target interaction matrix, the fused drug matrix and the fused target matrix. Here, we define the drug-target interaction as a binary matrix $A \in \{0,1\}^{N\times M}$, N represents the number of drugs, and M represents the number of targets. If $A_{ij} = 1$, it shows that there is a known interaction between the i-th drug and the j-th target, otherwise $A_{ij} = 0$. The relationship between the drugs is expressed as drug matrix U with U_{ij} (the i-th drug and the j-th drug), and the relationship between the targets is expressed as a target matrix P with P_{ij} (the i-th target and the j-th target), and U and P are normalized by (3) and (4), respectively:

$$U = {D_d}^{-\frac{1}{2}} U^d {D_d}^{-\frac{1}{2}} \tag{3}$$

$$P = {D_p}^{-\frac{1}{2}} P^p {D_p}^{-\frac{1}{2}} \tag{4}$$

D_d is the degree matrix of the drug, D_p is the degree matrix of the target, The definitions of D_d and D_p are as follows:

$$D_d = diag\left(\sum_j U_{ij}^d\right) \tag{5}$$

$$D_p = diag\left(\sum_j P_{ij}^p\right) \tag{6}$$

The constructed heterogeneous network matrix G is defined as follows:

$$G = \begin{pmatrix} U & A \\ A^T & P \end{pmatrix} \tag{7}$$

2.5. Encoder

GCN is a neural network structure with multiple connections. Based on the obtained feature representation before, we used the GCN encoder to learn the low-dimensional representation of the node from the graph.

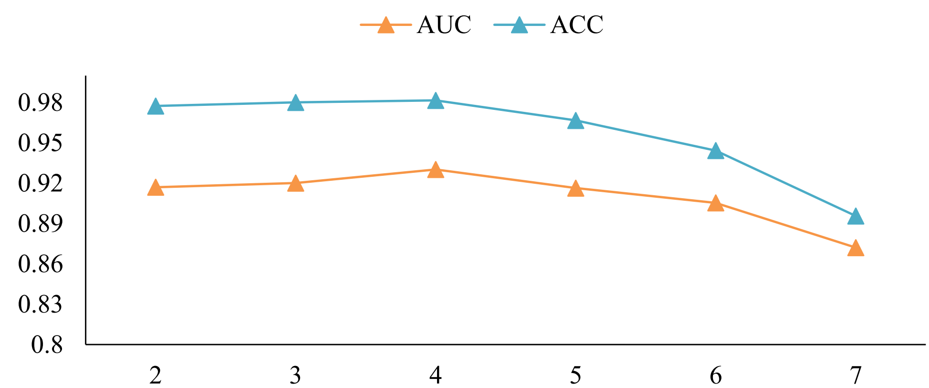

Fig. 2. The effect of the number of network layers on the prediction results.

Each layer of GCN aggregates the neighbor information to reconstruct the embedding as the input of the next layer based on the direct link of graph.

Given the adjacency matrix G of the graph, the propagation rule of the graph convolutional neural network is defined as:

$$X^{(l+1)} = f\left(X^{(l)}, G\right) = \sigma\left(D^{-\frac{1}{2}} G D^{-\frac{1}{2}} X^{(l)} W^{(l)}\right) \tag{8}$$

Here $X^{(l)}$ represents the input of the l-th layer, and $X^{(l+1)}$ represents the output of the l-th layer, that is, the input of the $(l+1)$-th layer. σ is the nonlinear activation function, D is the degree matrix of G, and $W^{(l)}$ is the trainable weight matrix.

The input $X^{(0)}$ of layer 0 is:

$$X^{(0)} = \begin{pmatrix} 0 & A \\ A^T & 0 \end{pmatrix} \tag{9}$$

The embedding from multiple convolution layers reflects the proximities of different orders between nodes in the network [31], therefore we resort to the attention mechanism [32] to integrate all useful structural information from multiple graph convolution layers.

After L iterations, we have got a low-dimensional embedding with dimension k. In our experiment, k is a hyperparameter, the prediction effect is optimal when k is set to 64.

2.6. Attention mechanism

The GAT (Graph Attention Network) [33] uses the attention mechanism to weight and sum the features of neighboring nodes. The weight of features of neighboring nodes depends entirely on the features of the nodes and is independent of the graph structure. In essence, GAT just replaces the original GCN standardization function with a neighbour node feature aggregation function using attention weights. Different from the attention mechanism between nodes, here, we introduce the attention mechanism between graph convolutional layers. The attention mechanism between the graph convolutional layers is a component of the graph neural network coding layer, which is responsible for allocating the inter-dependence between the graph convolutional layers. The output X of encoder is:

$$X = \sum_{i}^{n} a_i X^i \tag{10}$$

Here a_i is the attention parameter of the i-th layer and X^i is the output of the i-th layer.

2.7. Decoder

In order to reconstruct the drug-target interaction matrix, we built a decoder by the references [34]:

$$A^* = sigmoid X^d W^* X^p \tag{11}$$

Here W^* is a trainable weight matrix, and X^d and X^p are the low-dimensional representation of the drug and the low-dimensional representation of the target calculated by the coding layer, respectively.

The output matrix A^* has the same dimension as the original drug-target interaction matrix A. All drug-target pairs with the value of 0 in matrix A will be assigned a prediction score through the decoder. The higher prediction score indicates the drug-target pairs have the great probability of interaction.

2.8. Loss function

From a dataset with N drugs and M targets, we take drug-target association pairs as positive instances and take other pairs as negative instances. Differentiating two types of drug-target pairs is a binary classification problem. The number of associations is much less than that of the drug-target pairs, there are many undiscovered associations in all drug target pairs. Here, we adopt the weighted cross-entropy as loss function:

$$Loss = -\frac{1}{N \times M}\left(\lambda \times \sum_{a_{ij}^{'} \in y^+} log\left(a_{ij}^{'}\right) + \sum_{a_{ij}^{'} \in y^-} log\left(1 - a_{ij}^{'}\right)\right) \tag{12}$$

where a'_{ij} represents the prediction result of drug i and target j, y^+ and y^- represent the number of positive and negative samples, respectively, $\lambda = |y^+| / |y^-|$, the weight factor λ imposes the importance of observed associations to reduce the influence of data imbalance. N and M represent the number of drugs and targets, respectively.

3. Results

3.1. Hyperparameter

The choice of hyperparameters significantly affects the performance of the model. In this part, we discuss the hyperparameters set in the experiment. There are several main hyperparameters in our model, which are the feature embedding dimension k, the number of layers of the graph convolutional network L, the learning rate lr of the optimizer, two dropout rates (node dropout and regular dropout) β and γ, training epoch α.

The graph convolutional network needs to consider the over-smoothing phenomenon when setting the number of network layers, and the over-smoothing phenomenon is the main reason why the graph convolutional network cannot improve the model performance by superimposing the number of network layers like a neural network. In those experiments, we found when the number of network layers is set to four layers, the experimental effect is optimal. After the number of network layers exceeds four layers, both AUC and ACC have significantly decreased. Fig. 2 shows the effect of the number of network layers on the prediction results. Some researches point out that this is because

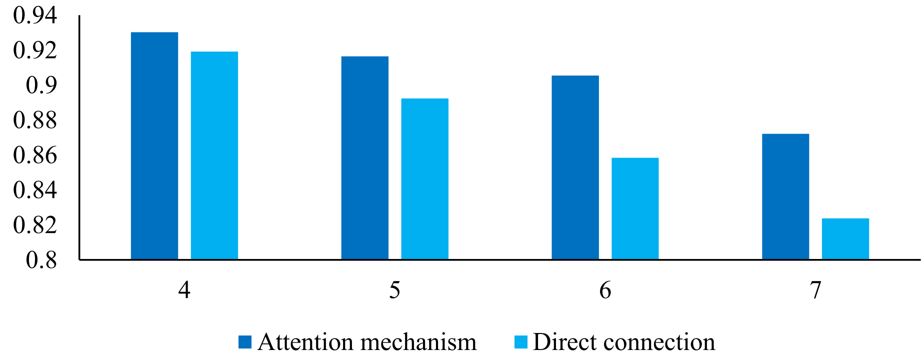

Fig. 3. The effect of adding attention mechanism on AUC with different network layers.

the over-smoothing phenomenon of the graph convolutional network will cause the values of the local features to converge to the same value [35], resulting in a decrease in the performance of the graph convolutional network. ResGCN-28 [36] used residuals, dense links, and expanded convolution to solve the problem of gradient disappearance and over-smoothing in some networks of graph convolutional networks, but there is no effective solution in link prediction, node classification and other fields.

The learning rate of the optimizer has a significant impact on the graph convolutional network. When the learning rate is larger, the running time of the graph convolutional network is shorter, but the loss function may not converge. If the learning rate is small, the convergence speed of the loss function is slower, and the running time of the graph convolutional network will be longer. We employ Adam optimizer [37] to minimize the loss function. Adam optimizer can update the weights of neural network iteratively based on training data. In order to prevent over-fitting, we also introduce a random drop strategy in the graph convolutional layer, including node dropout and regular dropout. By adjusting the parameters empirically, we set the feature embedding dimension to 64, the number of network layers to 4, the learning rate to 0.01, the node dropout to 0.6, the regular dropout to 0.6, and the epoch to 4000.

We used the attention mechanism on models with 4, 5, 6, and 7 graph convolutional network layers. Compared with the model in which each graph convolutional layer is directly connected, the performance of the model with the attention mechanism is better. Fig. 3 shows the performance of the model with the attention mechanism and the directly connected model. The above experiments prove the effectiveness of the attention mechanism in GCHN-DTI.

3.2. Performance evaluation

Different validation methods have been used for measuring the model performance. In our experiments, we adopt 5-fold cross-validation to evaluate the performances of prediction methods. In 5-fold cross-validation, all known drug-target associations are randomly split into five equal-sized subsets. The cross-validation process is repeated five times, and every subset is used as the testing set in turn while the remaining four subsets are used as the training set. In each fold, the prediction model is constructed on known associations in the training set and is used to predict associations in the testing set. For each prediction model, in each fold cross-validation, we considered the following evaluation metrics: Based on the methods scores, we calculated true positive (TP), false negative (FN), false positive (FP) and true negative (TN), accuracy (ACC), recall and specificity values as follows:

$$ACC = \frac{TP + TN}{TP + FP + TN + FN} \tag{13}$$

$$Recall = \frac{TP}{TP + FN} \tag{14}$$

$$Specificity = \frac{TN}{TN + FP} \tag{15}$$

3.3. Result

In this part, we compare GCHN-DTI with deepDTINet [38], DTINet, DDR, DNILMF [39] and NRLMF [40]. Fig. 4 shows the results of comparison with these methods. GCHN-DTI achieves the best AUC (AUC = 0.923), which improves over DDR, deepDTINet, DTINet, DTINET, DNILMF and NRLMF by 1.2 %, 1.7 %, 3.3 %, 4.4 %, 8.9 % and 8.9 %, respectively. Note that GCHN-DTI outperforms deepDTINet, DNILMF, NRLMF on AUPR, but is slightly worse than DDR and DTINet on AUPR. The GCHN-DTI method showed better performance, achieving an AUPR of 0.376, which is superior to that of deepDTINet, DNILMF, NRLMF by 0.4 %, 16.6 % and 10.6 %, respectively. In order to prove the

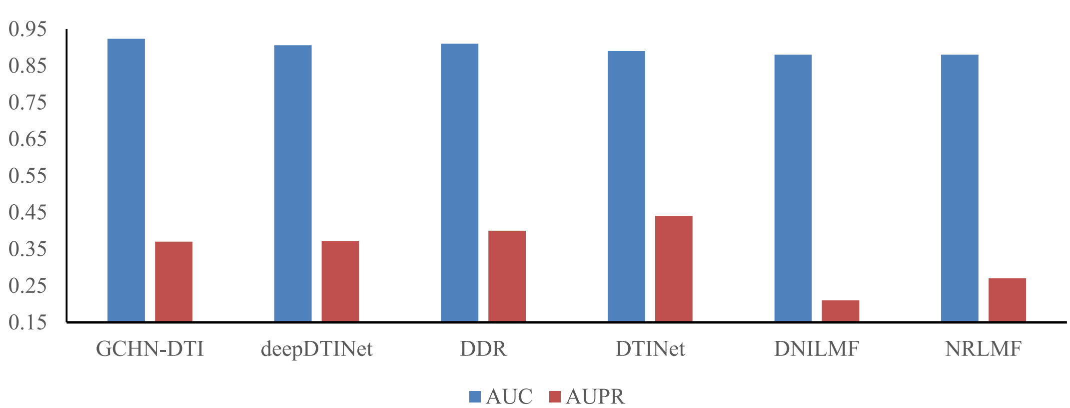

Fig. 4. Comparing AUC and AUPR performance of our method with other methods, and GCHN-DTI achieved the best AUC.

Table 3
Number of interacting nodes in Dataset1.

	ACC	Recall	Specificity	AUC
Interaction network	0.9250	0.7187	0.9251	0.8784
Similarity network	0.9318	0.7113	0.9318	0.8845
Fusion network	0.9831	0.7604	0.9832	0.9235

Table 4
Fivefold cross-validation results on an independent test set.

	AUC	AUPR	ACC	recall	specificity
1	0.9794	0.5813	0.9835	0.8780	0.9842
2	0.9640	0.5404	0.9805	0.8407	0.9815
3	0.9753	0.5492	0.9816	0.8441	0.9826
4	0.9744	0.5769	0.9830	0.8711	0.9838
5	0.9795	0.5883	0.9827	0.9054	0.9833
AVE	0.9745	0.5673	0.9823	0.8678	0.9831

effectiveness of network fusion, we set up a set of controlled trials. The group without network fusion uses only the drug similarity network, protein similarity network. But the group with network fusion fuses the two networks of drug similarity network, protein similarity network, drug-drug interactions and protein–protein interactions. Table 3 shows the results of these two sets of experiments. We found that ACC, Recall, Specificity, and AUC have been significantly improved by using network fusion in Table 3. We thought that this is because more heterogeneous networks contain more information about drug-target nodes, so as to more accurately predict the interaction between drugs and targets.

3.4. Independent test set

To further test the performance of GCHN-DTI, we have also studied the ability of GCHN-DTI to predict the association between drugs and targets on different datasets. We used dataset 2 as an independent test set, The independent test set contains 210 drugs, 204 targets, and 1476 drug targets interaction pairs. The details of Dataset2 are shown in Section 2.2.

By fivefold cross validation, the performance evaluation parameters of GCHN-DTI are reported in Table 4. It can be seen from the evaluation results that AUC of GCHN-DTI is 0.9745, AUPR is 0.5673. The prediction performance of GCHN on the independent test set is higher. This is because the proportion of positive samples in the independent test set is higher, and there is more known interaction information on the independent test set. The independent test results show that GCHN-DTI have good robustness.

3.5. Case study

To further prove the capability of the proposed model in a more realistic DTIs prediction scenario, we introduce the case study. Based on the case study, we can acquire five drug-target pairs with higher probability scores predicted by the prediction model on the main dataset, and also acquire five drug-target pairs with lower probability scores predicted. Then, we search for the relevant evidence from DrugBank. The DTIs contained in the used datasets were collected before 2017, thus, we can do verification by using newly updated DTIs in the above databases. The predicted interactions and corresponding supporting evidence are shown in Fig. 5.

We found the evidence for the majority of predicted interactions and no interactions, and we carried out further research on these predictions. For example, DB00252 (Phenytoin) is an anticonvulsant drug used in the prophylaxis and control of various types of seizures. Q12809 is Potassium voltage-gated channel subfamily H member 2. It was reported that Phenytoin can targeting NaV 1.5 sodium channels to reduce migration and invasion in metastatic breast cancer [41]. DB00273 (Topiramate) is an anticonvulsant drug used in the control of epilepsy and in the prophylaxis and treatment of migraines, DB00555 (Lamotrigine) is a phenyltriazine antiepileptic used to treat some types of epilepsy and bipolar

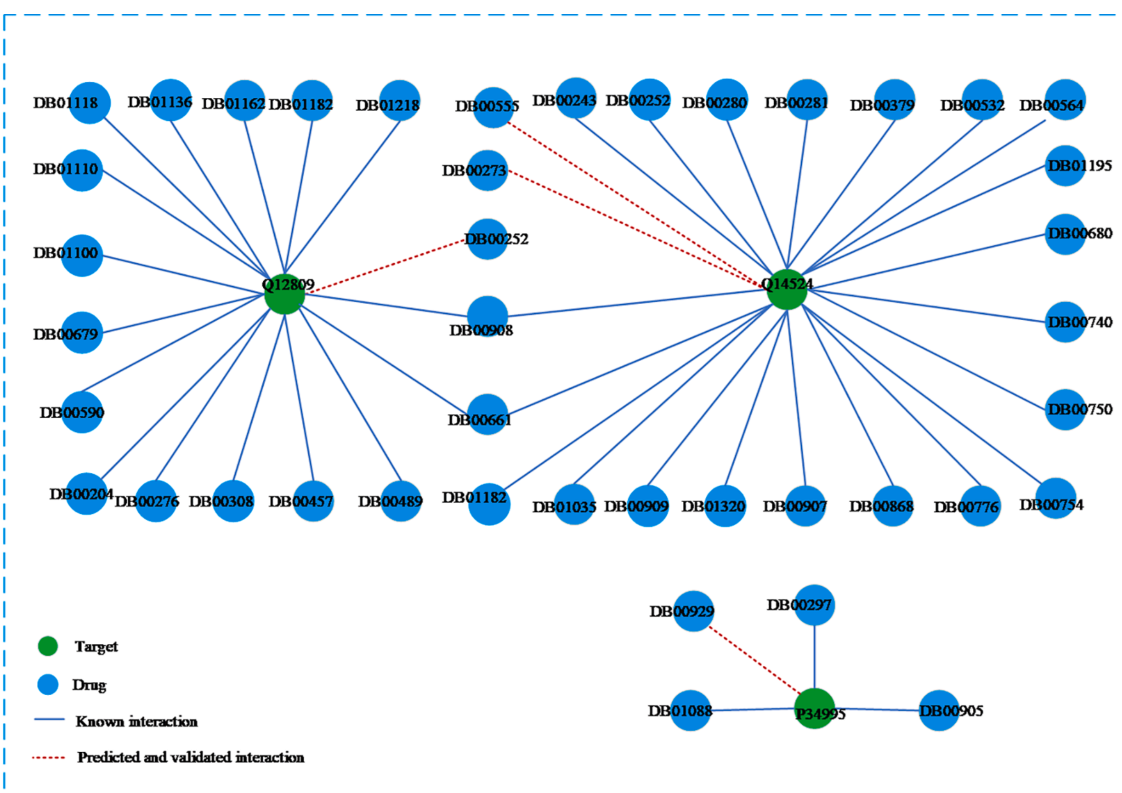

Fig. 5. The novel interactions predicted by GCHN-DTI.

I disorder, Q14524 is Sodium channel protein type 5 subunit alpha. In some research [42,43], DB00273 and DB00555 are proven to interact with Q14524. DB00929 (Misoprostol) can interact with P3499 (Prostaglandin E2 receptor EP1 subtype) as an Agonist. In addition, some drugs are predicted to have no interaction with the target, or it is generally intelligible that these drugs and targets have not still been found in research, and these evidences can be found in DrugBank. Therefore, the above results show that GCHN-DTI has a practical reference value for predicting potential drug-target interactions.

4. Conclusion

In this study, we established GCHN-DTI to identify potential drug-target interactions. GCHN-DTI is a heterogeneous network, and the graph convolution operation was performed on the heterogeneous network. GCHN-DTI captures the topological information of a heterogeneous network constructed from drug-target interaction network, drug-drug network, drug similarities network, protein–protein network, protein similarities network. After fivefold cross-validation, our method overall achieves higher prediction accuracy than other methods, and successfully predicted potential drug-target interactions. In addition, the prediction performance of the model will be significantly improved on datasets with a higher proportion of positive samples. In conclusion, GCHN-DTI is robust in different data sets, which will become an effective method for predicting drug-target interactions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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